WT Experiment 3

Topic: ESP8266 RFID Attendance System with Google Sheets

Class	Name	Roll-no
D15A	Siddhant Sathe	50
D15A	Arnav Sawant	52
D15A	Pranav Titambe	60

Introduction:

The ESP8266 RFID Attendance System is an innovative solution designed to streamline attendance tracking by integrating Radio Frequency Identification (RFID) technology with the ESP8266 Wi-Fi module. This system offers a modern approach to monitoring attendance, ensuring accuracy and efficiency in various environments such as educational institutions and workplaces.

Key Features:

- **Automated Attendance Recording:** Utilizes RFID cards or tags to automatically log attendance, reducing the need for manual input and minimizing errors.
- **Real-Time Data Processing:** The ESP8266 module processes RFID data in real-time, allowing immediate updates to attendance records.
- **Web-Based Interface:** Attendance data can be accessed and managed through a web-based platform, providing convenience and flexibility for administrators.
- **Wi-Fi Connectivity:** The system leverages the ESP8266's Wi-Fi capabilities to transmit data to a central server or cloud-based database, facilitating remote access and management.

Working Principle:

· Initialization: The ESP8266 connects to a Wi-Fi network and initializes the RFID reader.

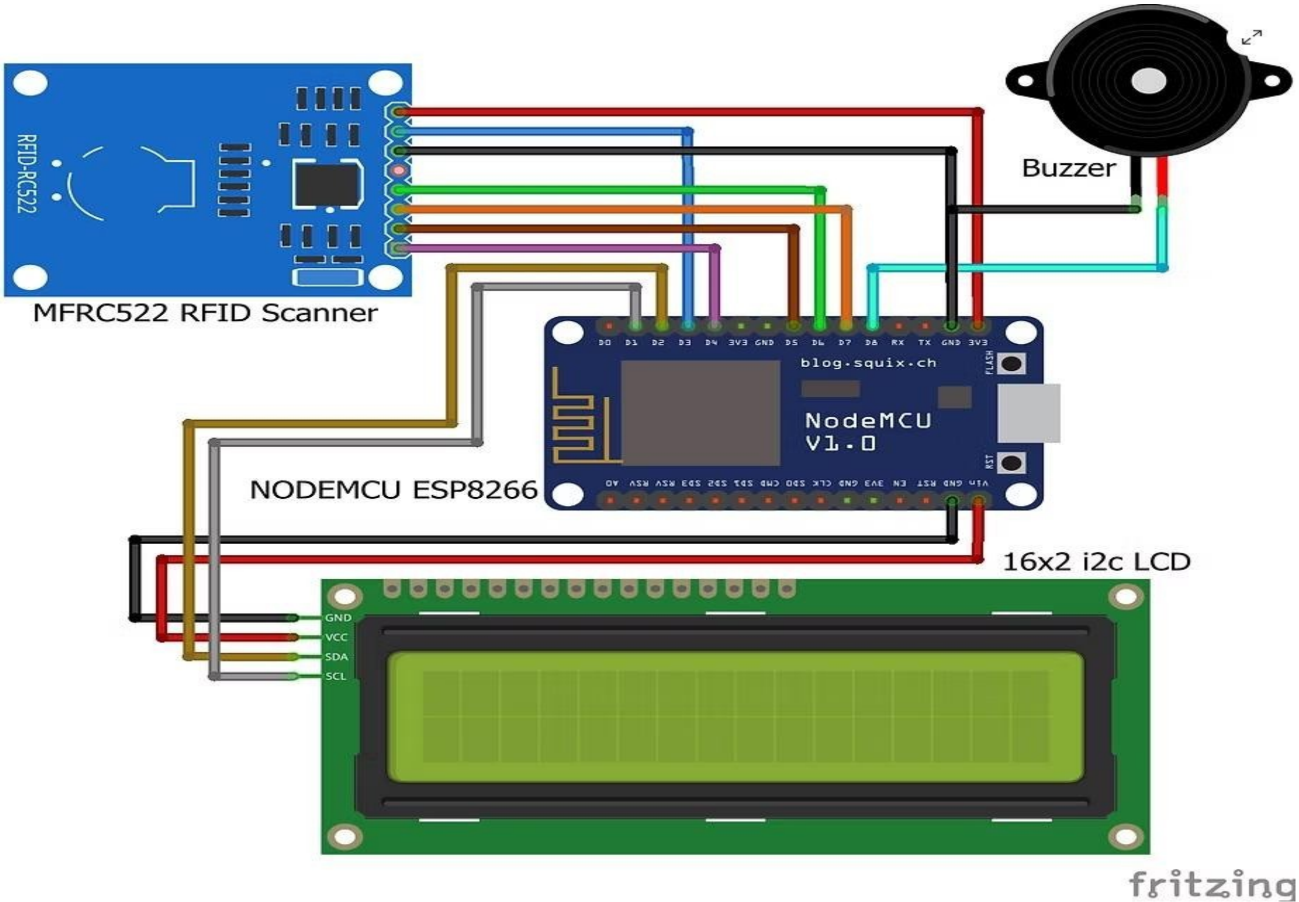
· **RFID Scanning:** When an RFID card/tag is presented to the reader, it captures the unique identifier (UID) of the card.

· **Data Processing:** The ESP8266 processes the UID, matches it with the corresponding user in the database, and logs the attendance with a timestamp.

· **Data Transmission:** The attendance record is transmitted via Wi-Fi to a central server or cloud-based platform for storage and further processing.

· **Feedback Mechanism:** Optional components like an LCD display can show confirmation messages, and a buzzer can provide audio feedback to indicate successful scanning.

Circuit Diagram:



Advantages:

· **Efficiency:** Automates the attendance process, saving time and reducing administrative workload.

· **Accuracy:** Minimizes human errors associated with manual attendance tracking.

· **Accessibility:** Enables remote access to attendance records through web-based interfaces.

· **Cost-Effective:** Utilizes affordable components, making it a budget-friendly solution for institutions.

· **Scalable:** Can be expanded to accommodate more users or integrated with other systems as need

Components Required:

- **ESP8266 (NodeMCU):** Serves as the main microcontroller, handling data processing and Wi-Fi communication.
- **RFID Reader (e.g., MFRC522):** Reads data from RFID cards or tags.
- **RFID Cards/Tags:** Assigned to individuals for identification purposes.
- **LCD Display (Optional):** Displays real-time information such as user identification and attendance status.
- **Buzzer (Optional):** Provides audio feedback upon successful scanning of RFID tags.
- **Power Supply:** Typically a 3.3V source to power the ESP8266 and associated components.

Conclusion:

The ESP8266 RFID Attendance System offers a modern and efficient approach to attendance tracking by combining RFID technology with Wi-Fi connectivity. Its real-time processing capabilities, ease of use, and scalability make it an ideal solution for various organizations seeking to enhance their attendance management processes.